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Empowering text classification with NLP and explainable AI for enhanced interpretability

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Abstract

Artificial intelligence (AI) models have demonstrated significant success in classifying various types of text. However, the complex nature of these models often complicates the interpretability of their classifications. To address these challenges and to enhance explainability, this study proposes a novel approach to text classification leveraging natural language processing (NLP) techniques and explainable AI (XAI) methods. Text preprocessing steps were essential for improving the quality of text analysis. This was gained by eliminating elements that contribute minimal semantic value. To achieve robust performance and mitigate the risk of overfitting, repeated stratified K-Fold cross-validation was utilized. Furthermore, the synthetic minority oversampling technique (SMOTE) was employed to address dataset imbalance issues. In the classification phase, nine machine learning models and hybrid/multi-model approaches were employed. To validate the explainability of the classifications, the local interpretable model-agnostic explanations (LIME) framework was utilized. The study utilized two datasets containing texts from domains such as sports, medicine, entertainment, politics, technology, and business. Empirical evaluations demonstrated the effectiveness of the proposed approach. The proposed hybrid model achieved exceptional performance across key metrics, including accuracy, precision, recall, and F1-score. The proposed hybrid model achieved results of up to 99% accuracy. This work can be used for various text analysis applications.

Keywords: Various text classification, Explainable AI, LIME, NTLK, Repeated stratified K-Fold, NLP

Introduction

Based on communication, linguistic features, structure, and style, texts are classified into some categories. Texts are divided into descriptive, narrative, expository, and persuasive types [1]. It can be understood by looking at the overall communication style and its key language features. This includes analyzing specific texts in detail. Functional labels, for instance, “Involved persuasion,” “Informational interaction,” and “Learned exposition,” are suggested for each type based on the analysis [2]. In this paper, some kinds of texts such as business, medical, technology, politics, entertainment, and sport are taken. They are identified using machine learning models and natural language processing (NLP).

Natural language processing (NLP) is a field that allows computers to manipulate and understand human language. It involves algorithms that generate texts that sound natural and analyze texts created by humans. It has gained significant attention because of advancements in computational power and research collaboration. Researchers in the area of NLP are focusing on extracting significant and statistical information from languages. They aim to apply these in diverse areas to improve interactions between machines and [3–5].

Text classification is a key aspect of natural language processing (NLP), and it has become an increasingly important area of study. Recently, it has been applied across some fields, for example, entertainment, technology, and medicine. Artificial intelligence techniques have been widely adopted to professionally categorize text into various categories as a result of the rapid growth of textual data. However, manual classification is costly, prone to errors, and often time-consuming.

Due to these issues, numerous studies have been conducted to classify texts into various contexts. Artificial intelligence models have been widely used to achieve this goal. In the field of education, the LI author classified English instructional content using decision trees, support vector machine, K-nearest neighbor, and naive Bayesian [6]. Wei [7] developed the Chi-squared test feature selection (CHI) algorithm for English and Chinese text classification. They stated that their work may be used to teach Chinese and English languages.

In the medical domain, deep learning models, such as RNN (recurrent neural network), GRU (gated recurrent unit), LSTM (long short-term memory), and CNN (convolutional neural network), have been used to diagnose patients' diseases [8]. 1130 emergency notes have been labeled in hospitals using machine and deep learning models [9]. They stated that their study utilizing deep learning models resulted in an accuracy of 94.5% identified texts, which is higher than machine learning models.

Last but not least, models for various context classifications, such as patent text, spam identification, emotion analysis, purpose assessment, and news classification, are proposed using LSTM, CNN, and RNN techniques [10–12].

The aforementioned research has already filled some gaps in the area, but they have some limitations, such as their models accept only one type of dataset, they require a high level of computing power, and their model accuracy is low. To overcome these, we

Table 1 The proposed model after LIME interpretations

Machine learning models	Accuracy	Computing power (CPU and RAM usages)
Linear discriminant analysis (LDA) AdaBoost classifier	Between (43–51%)	Medium computing power is required
Decision trees classifier K-nearest neighbor classifier Multinomial naive Bayes classifier	Between (81–96%)	Less computing power is required
Extra tree classifier Random forest classifier	96%	less computing power is required
Gradient boosting Classifier Support vector machine	Between (96–97%)	very high computing power is required
The proposed hybrid models (ridge classifier and logistic regression) After employing NLTK)	99%	Less computing power is required

propose a text classification with NLP and explainable AI. By offering a comparative study, this work also proposes an effective hybrid model. As shown in Table 1, our model supports six different types of datasets, needs less level of computing power, and outweighs other models' accuracy.

The primary contributions of the proposed work are as follows:

1. LIME interpretation has been used to better comprehend the model's classification.
2. The NLTK (Natural Language Toolkit) library is used to preprocess the selected dataset efficiently, which is then balanced using the advanced oversampling approach SMOTE (synthetic minority oversampling technique).
3. Repeated stratified K-Fold cross-validation was used instead of the traditional train–test split approach to mitigate the risk of overfitting.
4. The chosen model is evaluated on many text kinds, which helps to produce a more realistic, high-performing, and promising model.

The other remaining sections of the work are structured as follows: Current research is reviewed in section two. In section three, we present all of our techniques and strategies. The discussion and results are provided in section four. The last section presents the conclusion and future work.

Literature review

Text categorization systems are becoming increasingly in demand. It is essential to the domains of education, entertainment, and healthcare. The field of text classification has benefited from several models developed by researchers. It uses a variety of machine and deep learning models and has been the subject of extensive research [13–15].

Machine and deep learning models

In 2018, Cai et al. used a deep learning model of bidirectional-LSTM, which captures bidirectional “ n -gram” information with variable length. It works well for short news classification, and their model was about 85% accurate in news text context recognition [10]. Kim and Jeong employed convolutional neural networks to categorize sentiment. As they stated, the model could categorize longer texts. Their model's accuracies were approximately 81% for the binary classification and 68% for the ternary classification [16]. BERT (bidirectional encoder representations from transformers), a deep learning model for natural language processing (NLP) developed by Google, was also employed by Gao and Feng for sentiment classification [17]. The model achieved an accuracy of 77%. It may be considered suboptimal for certain applications. BERT demonstrates significance for various text classifications; however, it needs high-performance computing systems, such as RAM and external GPU, to ensure efficient optimization and processing [17].

For the research in 2020 and onward, a multi-model of CNN LSTM and Bio-BERT were proposed for biomedical text mining and [18, 19]. The authors of the works mentioned that their works were done on high computing power but with adequate accuracy. Kuriyozov et al. managed text detection with an accuracy of 85% from various contexts using RNN, CNN, and BERT [11]. Cang et al. (2024) refined the ALBERT model by combining CNN, Bi-LSTM, and attention processes to attain an accuracy of roughly 91%

in medical text mining. Text classification is accomplished using both the self-attention mechanism (QSAM) and the quantum recurrent neural network (QRNN). According to Chen and Fang [20], their model's accuracy on the RP dataset was 96.77%. Last but not least, disease classification using ensemble machine learning has been reported with an accuracy of 99% by [21]. Sultana et al. [22] evaluated the random forest algorithm for thyroid disease detection, also achieving an accuracy of 99%. Furthermore, [23] selected deep learning architectures of CNN and B-LSTM for sentiment analysis, they gained an accuracy of 94%.

Explainable artificial intelligence (XAI)

Explainable artificial intelligence (XAI) is a set of methods and approaches that allow human users to comprehend and trust the results generated by machine learning algorithms without compromising their functionality [24, 25]. Because AI algorithms are unable to provide an explanation for their choices due to their black-box nature, explainable artificial intelligence (XAI) was created [26–28]. Some researchers have adopted this method to assist people in understanding the behavior of the models they use [29, 30].

Ling et al. [29] worked on text-based classification models with interpretable representations [29]. They developed machine learning models to classify temporal bone fractures by utilizing CT images of 164 temporal bones. The algorithms that they used are logistic regression, random forest, XGBoost, and support vector machine. They utilized the average word frequency score (WFS) to identify relevant keywords. Also, they highlighted the frequency differences between positive and negative classifications. LIME is used as well to demonstrate how specific words contributed to the categorization of bone fractures. The random forest model's interpretability is improved in news classification by Tabassoum et al. [30] through the XAI approach LIME [30]. To attain this, they provided crucial information on the phrases that had a major impact on the model's prediction. Moreover, they examined the connections between model predictions and disturbed instances. Guleria et al. [31] explore using ensemble classification models within the explainable artificial intelligence (XAI) framework to predict heart disease from cardiovascular datasets [31]. The dataset includes 303 instances and 14 attributes of various types (categorical, integer, real), with the task being classification. The classification methods used are support vector machine (SVM), AdaBoost, K-nearest neighbor (KNN), bagging, logistic regression (LR), and naive Bayes.

Previous research on text classification using machine and deep learning models has addressed some gaps in the field, but these models are limited to specific types of datasets, require significant computing power, and often have lower accuracy. This work aims to overcome these issues by improving text classification interpretability through NLP and XAI, while also enhancing model performance to achieve higher accuracy and support various text types. The study uses LIME [24–30] to better understand model complexity and predictions, comparing nine models: linear discriminant analysis (LDA), decision trees, support vector machine, K-nearest neighbor, multinomial naive Bayes classifier, random forest, extra tree, gradient boosting, AdaBoost, and a hybrid approach (ridge classifier and logistic regression). After evaluating their performance, the hybrid approach was selected as the optimal choice for text classification.

Research methodology

This research employed a range of tools, including the Python programming language and several libraries. NumPy was used for mathematical operations, Pandas for data manipulation and analysis, Matplotlib for generating diverse plots, LIME for the proposed model predictions enlightenment, and scikit-learn for tasks such as building machine learning models, data preprocessing, feature selection, and model evaluation. Moreover, this section sheds light on the model we used for our experiment. Figure 1 shows the study's architectural layout. Thus, the following actions are required for the experiment:

1. We chose 2494 samples from two datasets in the first step: general dataset (2226 samples) and medical dataset (268 samples). Then combined the two datasets to create a new one.
2. The chosen dataset is effectively preprocessed using the NLTK (Natural Language Toolkit) library in the second stage. The dataset is then balanced using SMOTE's advanced oversampling technique (synthetic minority oversampling technique). Then, repeated stratified K-Fold cross-validation was used instead of the traditional train–test split approach to mitigate the risk of overfitting.
3. The third phase involves applying different machine learning algorithms to the data that have been preprocessed.
4. The **Hybrid Model** of ridge classifier and logistic regression have been chosen as the optimal model after utilizing natural language processing (NLP). As they did very well in text classification among all the other used models.
5. Finally, the model's operation and classification are then better understood by applying LIME explainer.

Dataset description

This study combined two datasets with six target classes, which is shown in Table 2.

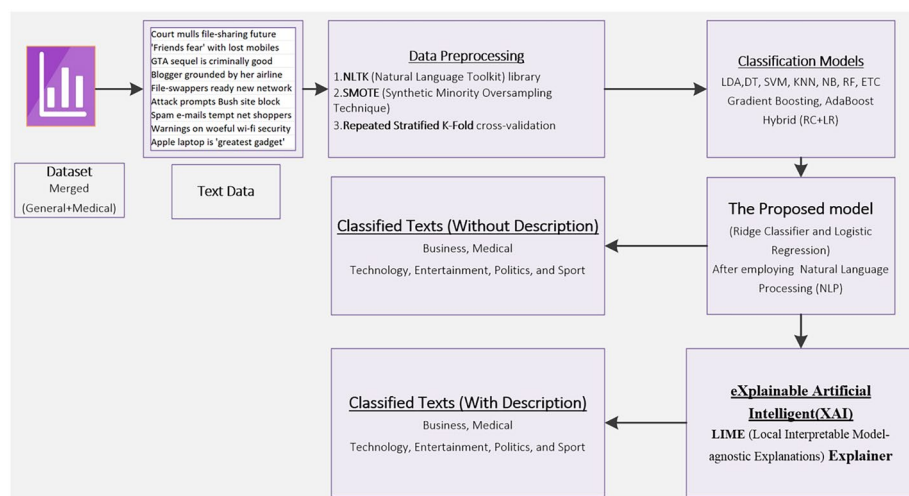


Fig. 1 The architecture of the study

Table 2 The number of samples for target classes of the used datasets

Datasets	Target class	No of samples	Download source
Medical dataset	Medical	268	https://github.com/McGill-NLP/medal/blob/master/toy_data/sample.csv
General dataset	Business	508	https://www.kaggle.com/code/sharatsk/nlp-text-document/input?select=df_file.csv
	Technology	401	
	Entertainment	387	
	Politics	418	
	Sport	512	
	Total (2226) samples		
Combined datasets	(Medical)	(268)	
	(Business	(2226)	
	Technology	Total (2494) samples	
	Entertainment		
	Politics		
	Sport)		

Table 3 The effectiveness of using NTLK Library

Text	Predicted without NTLK Library
Weak data buffets French economy A batch of down-beat government data has cast doubt over the French economy's future prospects...	Technology text—Wrong Class
Sky takes over Oscar night mantle Sky has signed a major new deal to broadcast this year's Academy Awards, taking over from three years of live Oscar coverage on the BBC	Technology text—Wrong Class
Texts	Predicted with NTLK Library used the following steps
	Text Tokenization
	Removing Stop words (Articles: the, a, an Prepositions: in, on, at, with, for Conjunctions: and, or, but, if Pronouns: he, she, it, we, they Auxiliary Verbs: is, are, was, were Other common words: this, that, of, to, by, for, from)
	Removing Punctuations ("!"#\$%&'()*+,-./:;<=>?@[\\]^_`{ }~")
Weak data buffets French economy A batch of down-beat government data has cast doubt over the French economy's future prospects...	Business text—Right Class
Sky takes over Oscar night mantle Sky has signed a major new deal to broadcast this year's Academy Awards, taking over from three years of live Oscar coverage on the BBC	Entertainment text—Right Class

Data preprocessing

NLTK (Natural Language Toolkit) library

NLTK is one of the first Python-based NLP development tools that was created by Dr. Edward Loper and Prof. Steven Bird. It includes statistically based text processing libraries for five core NLP supporting technologies as well as fundamental tools for semantic reasoning, such as text categorization, POS tagging, word tokenization, stemming, and semantic analysis [5, 32, 33].

The steps for preprocessing are shown in Fig. 1. We used NLTK because it helped improve the accuracy of our work. Some words, like “the,” “a,” and “an,” were removed, but these do matter for classification. Table 3 shows the effectiveness of using NLTK in

text classification. First, we break the text into smaller pieces (tokenization) to make it easier to analyze. Next, we remove unnecessary words (stop words). Finally, we removed punctuation and repeated stratified K-Fold cross-validation was used instead of the traditional train–test split approach to mitigate the risk of overfitting. Then, the SMOTE (synthetic minority oversampling technique) data balancing technique is applied to the cleaned texts.

SMOTE (synthetic minority oversampling technique)

Table 2 shows the imbalanced classes in the dataset. The imbalanced data increase training time and reduce performance. To solve this, the SMOTE (synthetic minority oversampling technique) method for balancing the minority class is leveraged among numerous tested methods for balancing [34–36]. In here SMOTE will treat with the highest category of sport with 512 samples. Synthetic samples are generated for other categories until they reach to sport samples which are shown in Fig. 2. If we also compare this to under-sampling techniques, this works crucial without information loss [35, 36].

LIME (local interpretable model-agnostic explanations) explainer

LIME is used to provide context for specific predictions made by black-box machine learning models. When you feed the machine learning model different data, LIME examines the results of the predictions. The altered samples and matching black-box model predictions are combined to create a new dataset via LIME. LIME then trains an interpretable model on this new dataset, which is weighted by how close the sampled examples are to the instance of interest [24, 26, 37, 38].

It can be expressed mathematically in Eq. 1.

$$\text{explanation}(x) = \arg \min g \in GL(f, g, \pi x) + \Omega(g) \quad (1)$$

LIME representation [38]

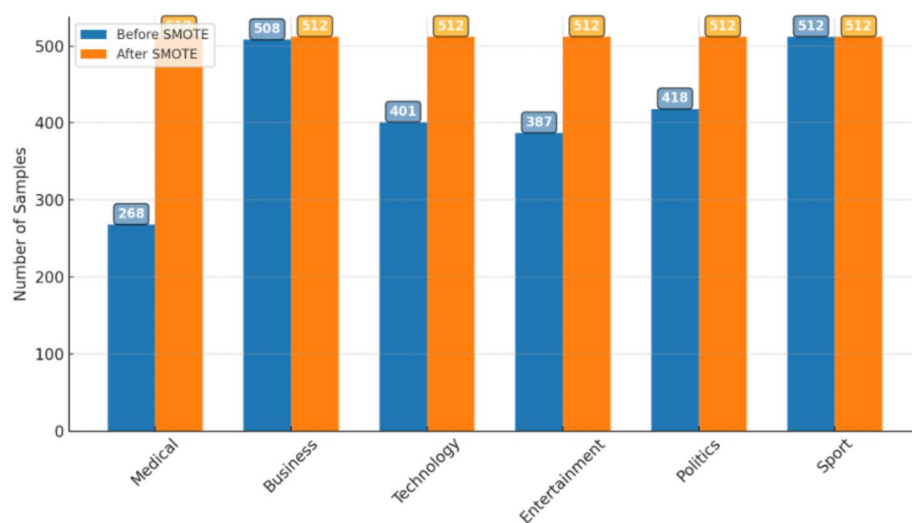


Fig. 2 The class distribution before and after utilized SMOTE

x is an instance. g is the explanation model that minimizes loss L (e.g., mean squared error), f is the original model (e.g., the **Hybrid** model). $\Omega(g)$ is model complexity (e.g., prefer fewer features). G is the family of possible explanations. πx is the proximity measure that defines how large the neighborhood around the instance.

LIME operates as follows as explained by Molnar [38]:

1. Identify the instance you want to explain within the context of a black-box model.
2. Generate variations of the dataset (perturbations) and obtain predictions from the black box for these new data points.
3. Assign weights to these perturbed samples based on their similarity to the original instance.
4. Train a locally weighted, interpretable model on the perturbed dataset.
5. Use the interpretable model to explain the black box's prediction for the selected instance.

The proposed model

The evaluation of ten machine learning models, detailed in Table 1, was conducted within the Jupyter Notebook Python environment on an MSI laptop equipped with a 9th-generation Intel Core i7 processor, 1TB hard disk, 32GB of RAM, and an NVIDIA GTX graphics card with 6GB of memory. The hybrid model—integrating logistic regression and ridge classifier algorithms—was selected. This model demonstrated superior classification accuracy while maintaining computational efficiency compared to the other models analyzed. Lastly, the model operations and predictions are then explained using local interpretable model-agnostic explanations (LIME) explainer.

The proposed hybrid models logistic regression (LR) and ridge classifier (RC)

Logistic regression (LR) Logistic regression (LR) is a fundamental supervised learning algorithm commonly employed for classification tasks where the output label is categorical [39–41]. It models the relationship between input features and the probability of a specific class outcome through a logistic function, as represented in Eq. 2:

$$\hat{y} = \frac{1}{1 + e^{-(\beta_1 x_1 - \beta_2)}} \quad (2)$$

Logistic Regression (LR) [42]

$\hat{y}$ is the predicted probability. x_i denotes the input feature data. e represents the base of the natural logarithm. β_1 and β_2 are the coefficients obtained during model fitting.

Ridge classifier (RC) Ridge classifier (RC) is a machine learning model built on ridge regression that adapts to classification tasks. It combines the advantages of linear models with regularization to avoid overfitting [43, 44].

It minimizes the following cost function as shown in Eq. 3:

$$\hat{y} = \arg \max_{c \in \{1, \dots, C\}} w_c^T x + b_c \quad (3)$$

Ridge Classifier (RC) [45]

$\hat{y}$ is the predicted class label. x is the input feature. C is the number of classes. w_c is a weight vector of class C and b_c is a constant offset added to the score for class c .

Assessment measures

We assessed the prediction model using many parameters including accuracy, precision, recall, and $F1$ -score [46]. The classifier performs better with higher precision and recall values. Accuracy relates to a classifier's overall effectiveness. The system's success is determined by the proportion of correct predictions in the total number of predictions, which is highly impacted by the dataset distribution. A model is regarded as good if it is highly accurate.

The following are the assessment criteria:

- $\text{accuracy} = (\text{True Positive} + \text{True Negative}) / (\text{Total Number})$
- $\text{precision} = \text{True Positive} / (\text{True Positive} + \text{False Positive})$
- $\text{recall} = \text{True Positive} / (\text{True Positive} + \text{False Negative})$
- $F1\text{-Score} = 2 / (1/\text{precision} + 1/\text{recall})$

Results and discussion

This section presents and discusses the results of the machine learning models, focusing on the impact of natural language processing (NLP) techniques.

Table 4 demonstrates the results of eleven machine learning models for text classification without employing NLTK and LIME interpretations.

The linear discriminant analysis (LDA) and AdaBoost classifier algorithms achieved the lowest fivefold Macro avg "Treats all classes equally unrelatedly of how many samples each class has" accuracy (MA%) for the AdaBoost model with medium computing power consumption using the combined datasets. But, weighted avg gives more importance to classes with more samples.

Compared to the LDA and AdaBoost models, the K-nearest neighbor and multinomial naive Bayes classifiers required less computational power while achieving better classification accuracy and faster processing times.

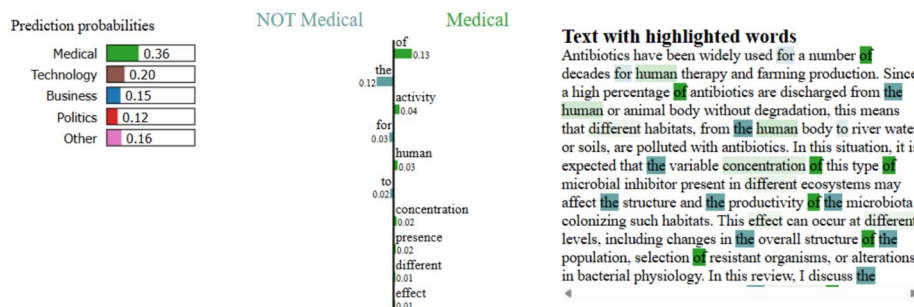
Finally, the support vector machine achieved an accuracy of approximately 0.97 but required significant computational resources. In contrast, the ridge classifier and logistic regression delivered an outstanding accuracy of 0.98 while utilizing fewer computational resources and generating results more quickly.

To gain deeper insights into the model's predictions, we employed LIME for interpretation. As a case study, we analyzed a medical text "*Antibiotics have been widely used for a number of decades for human therapy and farming production...*" sourced from the Internet to evaluate the model's predictive behavior.

As illustrated in Fig. 3, the model effectively identified the given text as medical and highlighted which specific words within the lengthy passage contributed to its classification as medical or non-medical. Notably, function words, such as articles, prepositions, auxiliary verbs, and punctuation, were not excluded from the text. However, these features, despite not being medical terms, were assigned the highest weights

Table 4 The results of evaluated machine learning models before employing NLTK

Models	Metrics	Precision	Recall	F1-Score
Linear Discriminant Analysis (LDA)	Accuracy			0.43
	Macro avg	0.45	0.47	0.41
	Weighted avg	0.45	0.43	0.39
AdaBoost Classifier	Accuracy			0.51
	Macro avg	0.57	0.54	0.55
	Weighted avg	0.53	0.54	0.51
Decision Trees Classifier	Accuracy			0.81
	Macro avg	0.82	0.82	0.82
	Weighted avg	0.81	0.81	0.81
K Nearest Neighbor Classifier	Accuracy			0.93
	Macro avg	0.94	0.93	0.93
	Weighted avg	0.93	0.93	0.93
Multinomial Naive Bayes classifier	Accuracy			0.96
	Macro avg	0.96	0.95	0.96
	Weighted avg	0.96	0.96	0.95
Random Forest Classifier	Accuracy			0.96
	Macro avg	0.96	0.96	0.96
	Weighted avg	0.96	0.96	0.96
Extra Tree Classifier	Accuracy			0.96
	Macro avg	0.97	0.96	0.96
	Weighted avg	0.96	0.96	0.96
Gradient Boosting Classifier	Accuracy			0.96
	Macro avg	0.96	0.96	0.96
	Weighted avg	0.96	0.96	0.96
Support Vector Machine	Accuracy			0.97
	Macro avg	0.98	0.97	0.97
	Weighted avg	0.97	0.97	0.97
Ridge Classifier and Logistic Regression	Accuracy			0.98
	Macro avg	0.98	0.98	0.98
	Weighted avg	0.98	0.98	0.98

**Fig. 3** The LIME interpretations for the model's classification in medical text

in the classification process. This inclusion led to increased computational time and a reduction in classification accuracy. The figure shows us how LIME effectively explained the predictions of the proposed model.

Figures 3 and 4 clearly explain how the models work and this led us to use NLTK for removing the stop words. After employing NLP the models we have used, attained promising cutting-edge results in terms of accuracy and less computing resource requirements.

Figures 3 and 4 provide a clear illustration of the models' functioning, prompting the decision to utilize NLTK for eliminating function words. As a result, the models

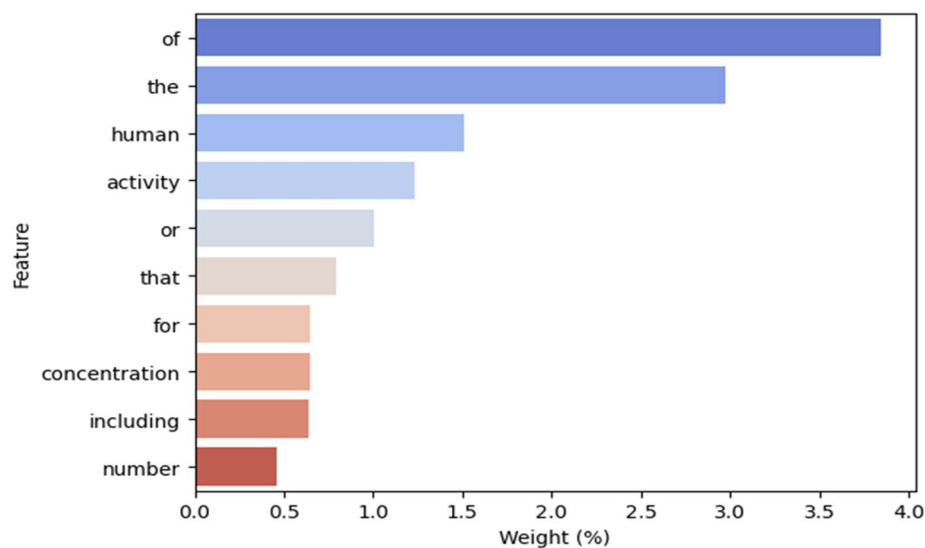


Fig. 4 10 features weight of the given text

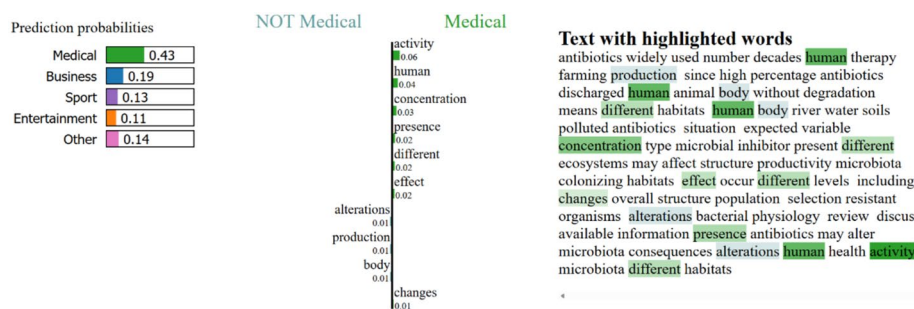


Fig. 5 The LIME interpretations for the model's classification in medical text after leveraging NLTK

achieved remarkable, state-of-the-art results, demonstrating high accuracy while minimizing computational resource requirements.

The classification accuracy for the medical text improved from 0.36, as shown in Fig. 3, to 0.43 in Fig. 5 after eliminating irrelevant words (features) that had been incorrectly identified as medical terms. This adjustment also altered the feature weight distribution, as depicted in Fig. 6, where only the most critical features were preserved.

Table 5 and Fig. 7 showcase the results of the evaluated models, demonstrating a notable enhancement in performance after using NTLK.

This leads to a dramatic improvement in the accuracy of certain models. For example, the accuracy of linear discriminant analysis (LDA) increased significantly from 0.43 to 0.64, while the AdaBoost classifier's accuracy improved from 0.51 to 0.59. However, other models, such as the extra tree classifier, gradient boosting classifier, and support vector machine, exhibited only marginal changes in accuracy (Fig. 8).

The confusion matrix shows that the proposed hybrid model performs with about 99% across all six categories. Most of all samples are correctly classified into their true categories, with only slight samples of them are misclassified. This reflects the model's ability to distinguish between topics with very high precision and recall.

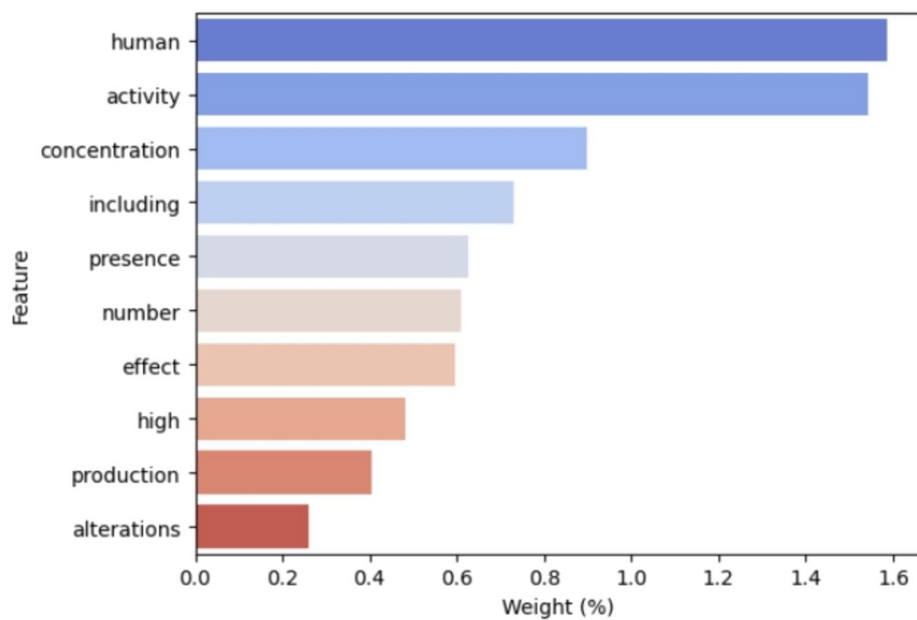


Fig. 6 10 features weight of the given text

Table 5 The results of evaluated machine learning models after employing NLTK

Models	Accuracy	Precision	Recall	F1-Score
AdaBoost classifier	0.58	0.62	0.58	0.59
Linear discriminant analysis (LDA)	0.65	0.66	0.65	0.64
Decision trees classifier	0.86	0.86	0.86	0.86
K-nearest neighbor classifier	0.94	0.95	0.95	0.95
Multinomial naive Bayes classifier	0.97	0.97	0.97	0.97
Gradient boosting classifier	0.97	0.97	0.97	0.97
Random forest classifier	0.98	0.98	0.98	0.98
Extra tree classifier	0.98	0.98	0.98	0.98
Support vector machine	0.98	0.98	0.98	0.98
The proposed hybrid models (ridge classifier and logistic regression) After employing natural language processing (NLP)	0.99	0.99	0.99	0.99

The presented hybrid models have promise for a variety of applications with an extraordinary accuracy of 0.99 and less use of computing resources. In medical text categorization, the proposed model improved illness detection and patient care. It also helps clinicians analyze clinical data and allows them to use it for more accurate diagnoses. In education, for example, in English language teaching, it can be used to categorize and analyze language patterns. It can be utilized for assessing writing skills and giving tailored feedback to improve learning outcomes and instructional tactics. Moreover, in the realm of emotion analysis, these models can identify and analyze emotional content in text. It is useful for applications like sentiment analysis, customer feedback evaluation, and mental health assessment. Thus, hybrid models can

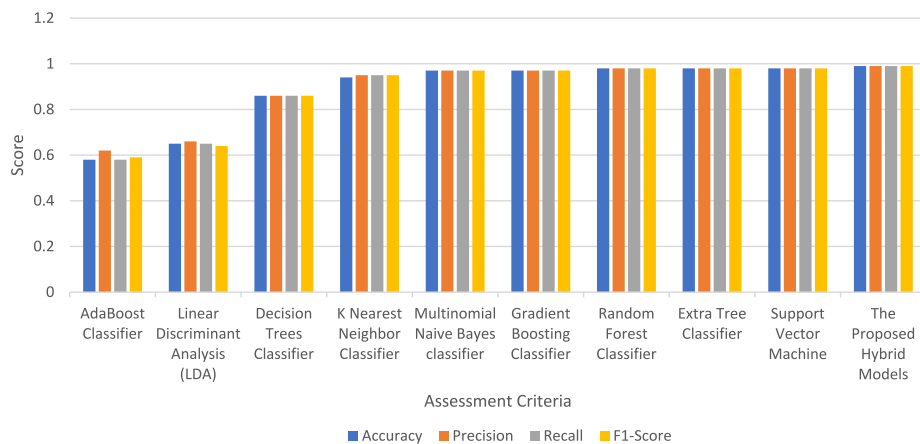


Fig. 7 The results of evaluated machine learning models after employing NLTK

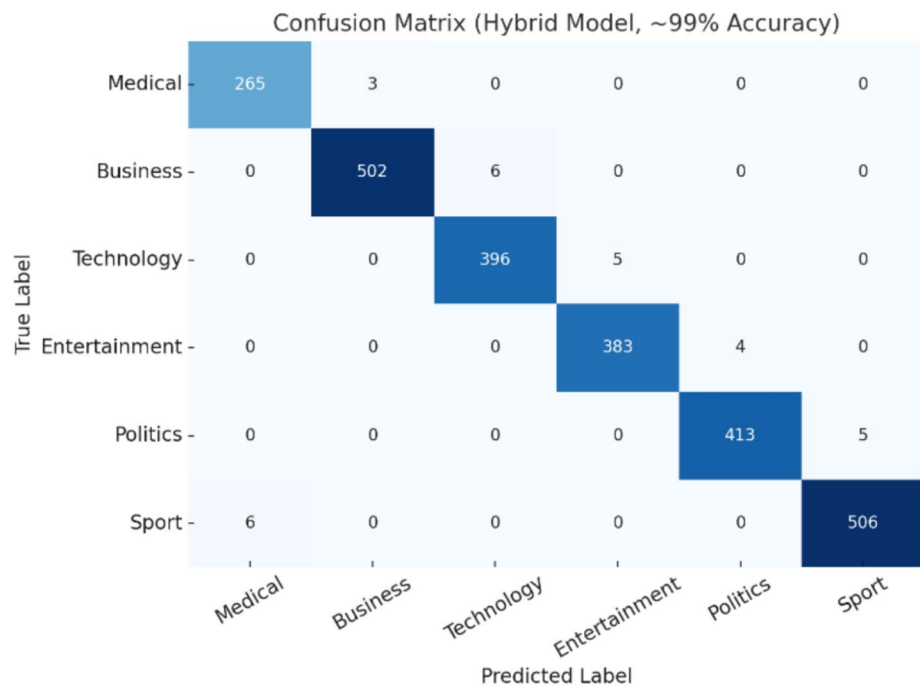


Fig. 8 The confusion matrix for the hybrid model with an accuracy of 99%

be used for a variety of areas, such as better communication, personalized experiences in healthcare, significant decision-making, emotional intelligence domains, and education.

Conclusion

This study explores the complexities of contemporary text classification using artificial intelligence. Its particular emphasis is on addressing the “black-box” nature of many models and the challenges posed by handling complex and potentially irrelevant data. To overcome these, the research proposes an innovative hybrid framework that merges the ridge classifier and logistic regression models. Our hybrid model was enhanced by

utilizing the NLTK of natural language processing (NLP) library. Additionally, it solves issues such as overfitting and class imbalance through the synthetic minority oversampling technique (SMOTE) and repeated stratified K-Fold cross-validation. Empirical evaluation revealed strong performance, with reported metrics up to 99% under specific conditions. These results highlight the robustness and adaptability of the hybrid framework across diverse application areas. Importantly, the study contributes to advancing AI-driven solutions in key fields such as healthcare, education, and emotional intelligence by enabling more transparent, communicative, and tailored applications.

Acknowledgements

We would like to acknowledge the reviewers and the journal that accepted and published our paper.

Author contributions

Sumaya contributed to the theoretical background, proposed the research title, and authored portions of the results and discussion sections. She also reviewed the entire paper. The practical aspects and the remaining sections of the paper were prepared and written by Mariwan.

Funding

Not received.

Data availability

Available on request.

Declarations

Competing interest

The authors declare that there are no competing interests.

Received: 26 June 2025 Accepted: 16 September 2025

Published online: 28 September 2025

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